FILING REF:

PROJECT PROPOSAL

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Project Title: AnchorAI

Group ID (As Enrolled in Canvas Class Groups): Gate 0

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Background/Aims/Objectives:

Gym members who stop attending cancel weeks or months later. That lag is what we target. The Retention Intelligence System segments members by behavior, predicts who will leave, and decides who to contact given a fixed budget and limited staff hours. Retention is the industry's biggest cost. One in three members leaves yearly, half of new joiners within six months, taking both dues and ancillary spend with them.

Current practice fails three ways:

- Calendar campaigns contact everyone, wasting budget on members who would have stayed.
- Threshold rules ignore trend. Twelve visits a month dropping to four beats a steady two, but rules treat them alike.
- Vendor scores give a number, not a decision. Staff cannot interrogate the flag, so they ignore it.

Our goals:

- Segment members interpretably using unsupervised learning.
- Predict churn with a calibrated gradient boosting and logistic regression ensemble.
- Turn predictions into a weekly action plan where every recommendation carries a challengeable justification.

Market Analysis:

The vendor landscape splits into two layers. The system of record, covering membership management, booking, payments and access control, is dominated by ABC Fitness including Glofox and Trainerize, alongside Mindbody, Xplor Technologies, Perfect Gym, Virtuagym and Daxko. Above it sits a thinner intelligence layer that bolts on rather than replacing the system of record. Keepme is the clearest example, best known for its predictive Keepme Score, targeting multi site operators and pricing by custom quote rather than published rates. Glofox offers a built in "At Risk" churn report. Current buyer guides for AI in gyms place Keepme at the head of the predictive retention and churn analytics category, with Replify leading AI reception and sales.

Four gaps define our opportunity:

- Competitors score risk but do not publicly expose constrained allocation, meaning the decision of which flagged members to contact and with which offer given a fixed budget and finite staff hours.
- Vendor messaging is uniformly built around targeting high risk members, despite academic evidence that risk ranking is suboptimal.
- Scores trained on tens of millions of profiles are powerful but unauditable, and mid market operators repeatedly cite staff distrust of black box flags.
- The intelligence layer is priced for larger chains. Keepme is generally recommended only once an operator reaches 5+ locations with real lead volume and a sales team to feed, leaving the one to five club operator, the bulk of the market by count, underserved.

We position this project as an explainable, budget aware retention planner for mid market chains, differentiating on transparent decision making rather than on model accuracy, where incumbents with far larger training corpora will always win.

Sources: *Health and Fitness Association 2025 Fitness Industry Benchmarking Report; Keepme and Glofox product documentation; industry buyer guides published in 2026.*
https://www.healthandfitness.org/health-fitness-association-releases-2025-fitness-industry-benchmarking-report/
https://www.keepme.ai/blog/making-the-most-of-your-data
https://www.glofox.com/blog/how-glofoxs-ai-can-mitigate-member-churn-and-boost-retention/
https://www.replify.ai/ai-sales-service-blog/best-ai-for-gyms-complete-buyers-guide

Project Description:

The system is scoped to cover four technique groups. Each group maps to a working component:

- **Decision automation:** A forward chaining rule engine encodes retention playbooks and commercial guardrails staff can read and edit. Fuzzy inference assigns engagement tiers, avoiding threshold cliffs. A case based reasoner picks the offer by matching similar past members and their outcomes.

- **Resource optimization:** Allocating a fixed budget across members, offers and channels is a constrained assignment problem, solved with a genetic

algorithm. Fitness is expected retained value net of cost, penalized for budget, staff hour, contact fatigue and inventory violations. An exact solver benchmarks small instances so we can report the optimality gap.

- **Knowledge discovery:** K means and Gaussian mixture segmentation, a calibrated churn ensemble, an uplift model for treatment effects, and class recommendations matched to member history.

- **Cognitive techniques:** A knowledge graph of members, clubs, classes, instructors and offers propagates cohort signals such as a cancelled class slot or departed trainer. A natural language assistant answers staff questions, grounded on the rule trace and graph.

Data Collection and Preparation

We will use a hybrid strategy. The public Kaggle dataset of 4,000 members benchmarks our pipeline, but it is a single month snapshot with no intervention data, so trajectory clustering and uplift modeling are impossible on it alone.

We therefore will also build a synthetic longitudinal simulator where each member is an agent with latent habit strength, motivation and social embeddedness, visiting via a process that decays during gaps and strengthens with attendance. It is calibrated to published benchmarks: roughly 66 percent annual retention, a first 90 day cliff, elevated January churn. Treatment effects are deliberately heterogeneous so the uplift model has real structure to recover. All parameters go in the appendix. Rule base and knowledge graph content come from two to four interviews with club managers or trainers, plus published effect sizes.

Dataset: Gym Customers Features and Churn, Adrian Vinueza, CC BY-NC-SA 4.0 https://www.kaggle.com/datasets/adrianvinueza/gym-customers-features-and-churn

Features are built as of a scoring date across recency, frequency, regularity, class engagement, relationship, commercial and lifecycle families. Trend and regularity carry the argument: absolute activity is a weak signal, deviation from a member's own pattern is a strong one.
Two preparation issues:

- **Leakage.** We exclude features generated by the cancellation process itself, such as freeze requests and payment method removal. These are the label in disguise, and likely explain part of the inflated accuracy in existing gym churn papers.
- **Censoring.** Members still active at window end are treated as unobserved, not silently coded as non churners.

Identifiers are pseudonymized, no health status is inferred, fairness is checked across age and gender, and contact fatigue caps are hard constraints in the optimizer.